

Skills Summary

Finding Inverses with Domain Restrictions

Use this reference while completing your worksheet or when studying.

Skill 1 — Inverting a measured model

Method. Write the rule as output = (expression in the input), then solve for the input. The result is the inverse rule.

Units swap with the variables. If the rule takes hours and returns dollars, the inverse takes dollars and returns hours — so answer in the inverse's unit, not the original's.

Example: A taxi charges $p(k) = 3k + 8$ dollars for k kilometres. A fare was 41 dollars. How far was the trip?

Step 1. The fare is known and the distance is wanted, which is backwards from the rule — so invert it rather than guessing values.

Step 2. $p = 3k + 8$ gives $k = \frac{p-8}{3}$, and $\frac{41-8}{3} = \frac{33}{3}$.
 $\Rightarrow k = 11$ km

Try it: A printer charges $p(n) = 4n + 6$ dollars for n posters. A bill was 46 dollars. How many posters?

check: $n = 10$

Skill 2 — Restricting the domain, then inverting

Why restrict. A rule with x^2 in it sends two inputs to the same output, so it has no inverse until one side is thrown away.

Recipe. Complete the square to find the vertex, restrict to one side of it, solve for x , and let the restriction choose the sign of the square root. The domain of f^{-1} is the range of the restricted f .

Example: $f(x) = (x - 2)^2 - 9$ on $x \geq 2$. Find $f^{-1}(16)$.

Step 1. The restriction starts at the vertex, so the inverse takes the plus square root — that decision has to be made before any arithmetic.

Step 2. $f^{-1}(x) = 2 + \sqrt{x+9}$, so $f^{-1}(16) = 2 + \sqrt{25} = 2 + 5$.
 $\Rightarrow f^{-1}(16) = 7$

Try it: $f(x) = (x - 5)^2 - 1$ on $x \geq 5$. Find $f^{-1}(35)$.

check: $f(11) = 35$

(!) **Watch out:** the minus square root is not wrong in general — it is the inverse of the *other* branch. Which one you keep is decided by the restriction, so write the restriction down first.

Skill 3 — Checking that two rules undo each other

Test. $g = f^{-1}$ exactly when $f(g(x)) = x$ and $g(f(x)) = x$ on the stated domains.

For a rational rule, substitute and then clear the compound fraction by multiplying the top and bottom by the inner denominator.

Example: $f(x) = 5x - 2$, $g(x) = \frac{x+2}{5}$. Show $(f \circ g)(x) = x$.

Step 1. Composing is the definition of “undoes”, so substitute g into f rather than comparing graphs.

Step 2. $5 \cdot \frac{x+2}{5} - 2 = (x+2) - 2$.
 $\Rightarrow (f \circ g)(x) = x$

Try it: $f(x) = \frac{7x+1}{x-2}$, $g(x) = \frac{2x+1}{x-7}$. Show $(f \circ g)(x) = x$.

$x = (x)(g \circ f)$:check